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BAGONG PILIPINAS

CC 101 – Introduction to Computing

Laboratory Activity No. 1 | Week 2: Introduction to Computing (CO1) | Unit 2, Computing Career Pathways

Explore Career Pathways

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Course/Program: BSCS 1-1	**Date:** September 16 2026

I. Objectives

At the end of this activity, you should be able to:

- Identify the major computing career pathways introduced in lecture and describe what each one involves.
- Choose one computing career pathway that best matches your interests and explain why.
- Map a realistic entry-to-expert roadmap for your chosen pathway, including representative job titles and key skills at each stage.
- Compare your current skills against your chosen pathway's requirements and identify a concrete plan to close the gap.

II. Materials Needed

- Your CC 101 Unit 2 lecture slides/notes on Computing Career Pathways
- This worksheet (printed copy, or the digital file to fill out and submit)
- Optional: internet access, for Part E (real job posting search)

III. Background

Computing is not just “programming.” As introduced in the Unit 2 lecture, the field branches into several distinct career pathways, each with its own operational scope, required technical competencies, representative job titles, and professional standard or reference body: Software Engineering, Systems & Cloud Infrastructure, Data Science & AI Engineering, Cybersecurity & Threat Defense, and Information Systems & Enterprise IT. Each pathway also has its own career roadmap — the entry-level, mid-level, senior-level, and expert/leadership roles a professional typically grows through. This activity asks you to review those five pathways, choose the one that interests you most, and map out what growing in that pathway could look like for you.

IV. Procedure

Part A – Recall the Five Pathways

1. Using your lecture notes or slides, complete Table 1 by writing the core operational scope and one to two representative job titles for each of the five computing career pathways.

Part B – Choose Your Pathway

2. From the five pathways, choose ONE that interests you the most.
3. In two to three sentences, explain why you chose it (e.g., skills you already enjoy, a subject you like, a job title that excites you).

Part C – Map Your Career Roadmap

4. If your chosen pathway is Software Engineering, Systems & Cloud Infrastructure, Data Science

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& AI Engineering, or Cybersecurity & Threat Defense, use the career roadmap shown for it in the lecture. If you chose Information Systems & Enterprise IT (no roadmap was shown in lecture), research typical entry-level, mid-level, senior-level, and expert/leadership roles for this pathway online.

5. Complete Table 2 with example job titles and key skills/responsibilities at each of the four career stages.

Part D – Skill Gap Self-Assessment

6. Looking at the essential technical competencies for your chosen pathway (Table 1 / lecture), list at least five in Table 3. For each, mark whether you already have some experience with it, and briefly note how you plan to learn it if you don't.

Part E – Real-World Job Search (Optional / Bonus)

7. Search a real job platform (e.g., LinkedIn, Indeed, JobStreet, Kalibrr) for one entry-level job posting that matches your chosen pathway. Record it in Table 4.

V. Data and Findings

Table 1. The Five Computing Career Pathways (Recap — Part A)

Pathway	Core Operational Scope	Representative Job Title(s)
Software Engineering	Designing, testing, and building applications and algorithms-driven software systems	Software Engineer, Backend Developer, Full-Stack Engineer
Systems & Cloud Infrastructure	Maintaining hardware provisioning, network virtualization, uptime, and deployment pipelines	Systems Administrator, DevOps Engineer, Site Reliability Engineer (SRE)
Data Science & AI Engineering	Extracting insights from structured/unstructured data and training predictive AI models	Data Scientist, Machine Learning Engineer, Data Engineer
Cybersecurity & Threat Defense	Safeguarding network perimeters, conducting penetration testing, and managing incident response	Information Security Analyst, Penetration Tester, SOC Specialist
Information Systems & Enterprise IT	Aligning business workflows with technology solutions and enterprise software architectures	Business Systems Analyst, IT Manager, Solution Architect

My Chosen Pathway (Part B)

I am most interested in: Software Engineer

Why I chose it:

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I chose Software Engineering because I've always been interested in computers and technology. I like learning how things work especially when it comes to apps, websites, nd games.

Table 2. My Career Roadmap for My Chosen Pathway (Part C)

Career Stage	Example Job Title(s)	Key Skills / Responsibilities
Entry-Level	Junior Coder	Mastering Best Practices , Algorithm Basics, Code & Debug, Learning Frameworks
Mid-Level	Mid-level Developer	Implementing Complex Features, Algorithm Efficiency, Unit Testing, Code Reviews
Senior-Level	Senior Engineer	System Architecture Design, Scalability & Performance, Mentoring Juniors, Full Stack Responsibility
Expert / Leadership	Technical Lead, Software Architect	Guiding Technical Decisions, Team Leadership, Architectural Overviews, Cross-functional Collaboration, High level System Design, Defining Standards, Integrating Complex Systems, Full Lifecycle Strategy

Table 3. Skill Gap Self-Assessment (Part D)

Essential Competency	Already Have It? (Y/N)	How I Plan to Learn It
Communication	Y	
Teamwork	Y	
Debugging	N	I will practice finding and fixing errors in my code
Following Instructions	Y	
Patience	Y	
Adaptability	N	I will try to adjust and learn whenever I encounter new software or technology

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Table 4. Real Job Posting Found (Part E, Optional / Bonus)

Item	Details
Job Title	
Company	
Platform / Source (e.g., LinkedIn, Indeed, JobStreet)	
Required Skills Listed (3–5)	

VI. Realization (Reflection)

In two to four sentences each, answer the following in your own words.

1. Why did you choose this particular pathway over the other four?

I chose Software Engineer over the other pathways because I'm more interested in computers, technology, and how software works. I feel like his pathway suits my interests and skills more, and I think I would enjoy learning and working in this field in the future

2. Looking at your roadmap, which career stage excites you the most, and why?

The career stage that excites me the most is becoming a Senior Engineer because I want to reach a point where I'm confident in my skills and can handle bigger projects. I also like the idea of gaining more experience, learning new things, and being able to help other developers along the way

3. Which required skill or competency for this pathway do you already have, and which one will be hardest for you to develop?

I already have is problem solving because I like figuring things out when I don't know how to do something. The hardest skill for me to develop would probably be math, because sometimes I find difficult to understand complicated problems, but I know it is important in Software Engineer, so I want to get better at it

4. How does mapping this roadmap change — or confirm — your plans for the rest of this course and your degree?

Looking the roadmap made me sure that software engineer is the path I want to take. I know there will be some subjects that I might struggle with, especially math, but I'll just do my best and focus on learning the things I'm interested in, like coding and technology

VII. Scoring Guide

Criteria	Description	Points

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Criteria	Description	Points
Completeness	Tables 1–3 and the reflection are fully filled out (Table 4 is bonus).	25
Roadmap Accuracy & Depth	Roadmap and skills recorded match the lecture pathway (or credible outside research for IS & Enterprise IT).	25
Realization / Reflection	Answers are thoughtful, specific to the student's own choice, and well written.	30
Neatness & Timeliness	Worksheet is legible/well-formatted and submitted on time.	20
Total		**100**

Checked by: **Score:** / 100

Instructor's Signature over Printed Name / Date

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